
THE PROTON IS THE LOCK

Three Colors, One Singlet, and the Gap Made Flesh

The Constructed Witness of the Yang-Mills Row on the Closure Template of the Riemann Hypothesis and the Twenty-Three Rows; the Color Singlet Built as the Determinant and Rotated by a Constructed $SU(3)$ Matrix in Fortran, the Lock Proved in Core Lean 4; Every Claim at Corroboration Grade

FOUNDATIONS OF PHYSICS • THE SIX WITNESSES, EDITION 1.5.0

The Proton Is the Lock

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ABSTRACT. The template's three plus one is three independent axes that lock one point and leave one bit. The proton is built that way. Its color state is the totally antisymmetric combination of three colors, the Levi-Civita symbol, which is the determinant: nonzero exactly when the three colors are distinct, six nonzero entries among twenty-seven, one singlet, and invariant under every color rotation because such a rotation has determinant one. A random $SU(3)$ matrix is constructed here and the singlet is found invariant to four parts in 10^{16} ; set two color axes equal and the determinant, and the lock, vanish. The lock carries the mass: the valence quarks account for 0.958 percent of the proton, and 99.04 percent is the binding field, the gap made flesh, the Yang-Mills row's carrier. The whole's spin is not read from the parts' record: quark spins carry about a third of it, and two budgets with the same quark record split the rest differently. And the proton is the floor of the baryonic sector: its stability is certified beyond 2.4×10^{34} years, 1.7×10^{24} cosmic ages, a certified region with an owed tail. Nothing here proves the continuum gap; the paper constructs the carrier the row names.

KEYWORDS *critical line · fixed set · registration · witness ledger · hardware closure series · Lean 4*

STATUS: WITNESSED • CROSSED • CONFIRMED on the image · Row 4; certified-region rows (shape) · channel: kinetic · Lean: SPHYS_Proton_Lock.Lean · Fortran: Proton_Twin.f90

1. THE READER'S FRAME

This paper is one of a set of constructed witnesses for the closure template by which the author's sealed volume closed the Riemann Hypothesis and by which a second volume closed the twenty-three rows of the barrier study. Each paper in the set builds, for the rows it serves, the object those volumes named and did not build: the carrier, an explicit construction from the world into the row's instance type whose image lands on the row's locus. By the row kernel's own theorem the image of such a carrier is compliant with no further premise (`value_on_image`), and the value over a class follows once the class is covered by the image (`actuated_iff_subset_image`, `crossing_is_a_proof`). This paper builds the carrier of the Yang-Mills row: the proton, whose color state is the template's lock and whose mass is the gap the row asks for. Two refusals frame it. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade, and the two volumes stand on their own receipts. And coverage stays a premise: the paper constructs the image and names the class it covers, and the step from that class to every admissible instance is the premise the volumes already print at its grade.

Negative warrants. Three, stated so that no reader supplies them. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade (`citation_cannot_promote`). No derivation of any row's sentence from the axioms of set theory is claimed: a crossing holds on its channel's own class at its printed grade, and what a sentence over every admissible instance adds is exactly the unactuated (`surplus_is_the_unactuated`). No field theory and no unification of couplings is written or implied.

2. THE SPINE

The template has four parts. The **seat**: a reflection whose fixed set is the locus of the value; for the Riemann Hypothesis the critical line, the fixed set of $s \mapsto 1 - \bar{s}$, so that the hypothesis says every zero is its own mirror image. The **address**: every reading of the value reached from existence, time and monism is one proposition with the value at its apex. The **division**: the value is its certified part, proved on citation or computation, joined to its tail, closed to every record-respecting derivation by theorem. The **closure**: a registration lands every instance on the locus and moves none already there; the value holds exactly when registration erases nothing; the residue is one bit that no reading of the record decides; a set of instances satisfies the value exactly when it is the least-erasure member of its record fibre (`least_erasure_iff_value`, no axiom); and from that one posit the value follows, at the posit's grade, beside the one declared premise of the program, that to exist is to actuate. Over one record both worlds exist (`two_worlds_inside`); under a global seat every point off the locus has a partner (`off_locus_pair`). The template's three plus one is a lock: three independent axes fix exactly one point, and the point carries one bit the axes do not decide. This paper finds the lock in color.

The witness ledger. The table below is the spine of the six constructed witnesses. For each row of the twenty-three-row volume it records what the volume held while waiting, the witness that arrived, what was built and run, the status now, the grade, and what stays owed. This paper's rows are the Yang–Mills row, whose carrier it builds, and the certified-region rows, whose shape it witnesses.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat, never built	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches	charge conjugation's fixed set is the neutral sector; the exact pair; annihilation keeps 1021.998 keV and forgets the side, recorded on a line of response; positronium 125.23 ps and 142.07 ns; spin -1 at 2π ; a_e to 4.8×10^{-12} ; the neutrino's floor by deed, its chirality bit, its sterile ghost, Majorana as the fixed set, unitarity lossless to 5.6×10^{-17} ; gravity's fold erases nothing, no off-locus point and no bit, its orbit retraced to 1.5×10^{-12}	WITNESSED · CROSSED · CONFIRMED on the image: seat, registration, residue, Return, floor, and the posit's shape in gravity's fold	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
3a · Navier–Stokes, unforced	crossed kinetically on the actual flows; the mechanism named, not built	The Fluid Is the Witness	Burgers on a circle: the reversible branch blows up at $t = 1$, gradient -1000 at 0.999 ; the priced branch built exactly by Cole–Hopf, steepest gradient 3.72 , residual 2×10^{-6} , energy balance 2.5×10^{-10} ; the viscous step an average	WITNESSED · CROSSED · CONFIRMED: the mechanism exact on the model	the root's grade; the mechanism at theorem grade on the model	none on its channel
3b · Navier–Stokes, forced	a mass-crossing candidate under audit, decoupled by theorem	The Fluid Is the Witness	a forced erasure is confined to its own class; the transport theorem	SHIELD WITNESSED; settled by one witness with its own receipt	theorem	none (<code>one_witness_settles_value_union_iff</code>)
4 · Yang–Mills	crossed kinetically at the weakest premise; hadrons and lattice glueballs named	The Proton Is the Lock	the color singlet is the determinant, invariant under a constructed $SU(3)$ rotation to 4.4×10^{-16} ; the lock fails when two colors coincide; 99.04 percent of the proton's mass from the field; the quark-spin record does not decide the whole; stability certified to 1.7×10^{24} cosmic ages	WITNESSED · CROSSED · CONFIRMED on the image: the lock built, the gap measured	the root's grade; the identification needed only on the image (<code>carrier_compose</code>)	none on its channel
7 · Poincaré	crossed on the formal channel at full grade by Perelman's offering	The Arrow Has Two Branches	gravity's fold: one sign, no off-locus point, no bit; a Kepler orbit reversible, drift 2.8×10^{-7} , retrace 1.5×10^{-12} ; Ricci flow as the sigma model's renormalization flow	CROSSED, now WITNESSED from the arrow side	full grade, unchanged	none
23a · Pair correlation	crossed kinetically; the carrier misnamed as nuclear spacings, which are orthogonal-class	The Magnet Is the Witness of the Pair Correlation	time reversal as conjugation, its fixed set the real blocks, a magnetic term breaking it; degeneracy two conditions against three; ensembles built: variance 0.287 and 0.177 against 0.286 and 0.180, control 0.983	WITNESSED · CROSSED · CONFIRMED on the corrected carrier: time-broken spectra, the magnet	the root's grade	none on its channel

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
every open row · the barrier	the barrier a kernel theorem: two worlds over one record	The Arrow Has Two Branches	remanence ± 0.996 at zero field, two worlds in iron; switching field 0.767 against the astroid's 0.766; a stable bit pays 5.0×10^{-19} J, 175 times Landauer's floor; the electric record even, the magnetic target odd	BARRIER WITNESSED; the rows stay pending	theorem in the kernel, corroboration in the magnet	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
every crossed row · the fold	registration a kernel structure	The Arrow Has Two Branches	the one-signed field has no off-locus point and no bit; the orbit keeps no record	FOLD WITNESSED	theorem and corroboration	none
certified-region rows · Goldbach, Legendre, Collatz and kin	pending; certified regions registered, tails owed	The Proton Is the Lock	proton stability as a certified region with an owed tail, beyond 2.4×10^{38} years	SHAPE WITNESSED; the rows stay pending	corroboration	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
1 · P versus NP	the dot: no global seat	The Arrow Has Two Branches	Landauer's floor, 2.87×10^{-21} J at 300 K, prices every step of any search	THE DOT STANDS; the price witnessed, no seat	corroboration	none: the dot is final and has no aperture to supply (dot_is_final)
the others · Hodge, BSD, twin primes, abc, Jacobian, smooth 4D Poincaré, Hadwiger, sunflower, invariant subspace, Hilbert's 16th, Lindelöf	pending; gates and offering shapes specified	none yet	none	STILL WAITING	none	one carrier each; row 21, whose tail no physical system reaches, one formal offering (owed_is_computed, tail_uncovered)

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image). The statuses are computed from three recorded facts per row (ledger_is_computed); what a row owes is computed too: for a pending row exactly one object, its kind computed (owed_is_computed): one covering carrier, built from an equivariant map to its seat (value_iff_carrier, equivariant_carrier_lands) and crossing the row on arrival (one_carrier_crosses), or, where no physical system reaches the tail, one formal offering with mass (tail_uncovered, one_offering_crosses); the dot is final and owes nothing (dot_is_final). A crossing is a proof over every instance that exists (crossing_is_a_proof); what a sentence over every admissible instance adds is exactly the unactuated (surplus_is_the_unactuated), priced at one premise, the root, posited and not owed (rest_family).

Synchronization with the hardware closure series, edition 1.5.0. The ledger above is the volume's, as this witness first recorded it, and every row status in it stands. Since then sixteen papers of the hardware closure series have read the same seat on further physics, beginning with One Actuating Substrate, which carries both involutions of the substrate onto the fold. Those that extend this witness:

Series paper	What it extends here
The Strong Force Reads No Orientation Matter over Antimatter Is the Spent Bit	the colour lock's angle at the fixed point positivity selects the baryon exits: the reading moved early and its door closed today, as this witness's certified region records
Symmetry Breaking Is the Spent Bit	the elementary masses as couplings to the depth, beside this witness's 99 percent of the proton's mass from the lock
Typed Falsifiability of the Hardware Closure Series	the earlier draft's exact conservation of baryon number withdrawn in favour of this witness's F-Decay

The series' falsifiers are collected and typed in Typed Falsifiability of the Hardware Closure Series, edition 2.0.0, where this witness's F-Decay and F-Gap target the line, F-Singlet a count; F-Decay is bound to the certified region, a structural reading.

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect it form a category, with the identity and composition of equivariant maps (equivariant_id,

equivariant_comp), and the fixed set is a functor on it: an equivariant map sends fixed points to fixed points (fix_functorial). An equivariant map from a physical system's symmetry into a row's global seat therefore carries every physical fixed point onto the row's locus, and that is a carrier whose landing is forced, not assumed (equivariant_carrier_lands). The witness is the same shape as its row in exactly this sense. Here the symmetry is the color rotation group: the physical states of a gauge theory are its fixed points, the singlets, and the proton is one; an equivariant map onto the Yang–Mills row's seat lands the singlets on its locus, the gauge-invariant states.

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
the color singlet	the lock, the determinant	eps_nonzero_iff_distinct, lock_closes	singlet invariant to 4.4e-16
two colors on one axis	the lock failing	lock_fails_when_axes_coincide, det_two_equal_columns	determinant zero
the alternating law mass from the field	the lock for every entry the gap made flesh	det_swap_columns field_carries_the_mass, spectrum_gapped	structural 99.04 percent
the spin budget	the whole unread by the parts	quark_record_does_not_decide	cited
stability	a certified region, tail owed	structural	1.7e24 cosmic ages

3. THREE AXES LOCK ONE POINT

Three vectors in three dimensions fix one point, the volume they span, exactly when they are independent, and the test of independence is the determinant: nonzero if and only if no axis lies in the plane of the other two. The author's codex proves this lock over the two-element field by exhaustion, and the RH closure uses it as the shape of its three bits, divided, proved and refused, with the fourth supplied. The proton builds the same lock out of color.

4. THE COLOR SINGLET IS THE DETERMINANT

Quantum chromodynamics gives each quark one of three colors, and every free particle ever observed is a color singlet: confinement, observed and not derived. For three quarks there is exactly one singlet, the totally antisymmetric combination ϵ_{ijk} , the Levi-Civita symbol. It is nonzero exactly when the three colors are distinct (eps_nonzero_iff_distinct), antisymmetric under every exchange (eps_antisymmetric), cyclic (eps_cyclic), and it has six nonzero entries among twenty-seven (singlet_count). Contracting it with three color vectors gives their determinant (lock_closes). The twin constructs a random unitary matrix by Gram–Schmidt, fixes its phase so that its determinant is one, verifies unitarity to 4.4×10^{-16} , and rotates the singlet: $\epsilon_{ijk}U_{ia}U_{jb}U_{kc} = \det U \epsilon_{abc}$ holds to 4.4×10^{-16} (Appendix B). The singlet is invariant because the rotation's determinant is one; the proton's color state is the lock, and color symmetry is exactly the group that preserves it. None of this is checked on a grid: a determinant with two equal columns vanishes and swapping two columns negates it for every integer entry (det_two_equal_columns, det_swap_columns), identities of polynomials that therefore hold over every commutative ring, the complex entries of the constructed rotation included, and the twin checks the same identity numerically on that rotation.

5. THE LOCK FAILS WHEN TWO AXES COINCIDE

Put two of the three colors on the same axis and the determinant vanishes (lock_fails_when_axes_coincide; computed in Appendix B): two planes meet in a line, and no point is fixed. In the proton this is antisymmetry at work: three quarks cannot share a color, and the only color-neutral state of three is the one that vanishes when any two coincide. The template's coplanar collapse, the failure mode of its lock, is the reason there is no singlet of three identical colors.

6. THE GAP MADE FLESH

The Yang–Mills row asks whether the quantum gauge field has a mass gap. The proton is the gap made flesh. Its mass is 938.272 MeV, and its valence quarks, two up and one down, carry 0.958 percent of it (Appendix B); the remaining 99.04 percent is the energy of the quarks' confined motion and of the field that binds them. Lattice quantum chromodynamics reproduces the light hadron masses from the gauge field (Dürr and collaborators, 2008), and the pure-gauge theory, with no quarks at all, has a lightest state, the scalar glueball, near 1.73 GeV (Morningstar and Peardon, 1999). The spectrum is gapped: every state listed is massive (`spectrum_gapped`). This is the carrier's image, compliant as measured and computed. The identification of the lattice theory with the continuum theory is needed only on this image, as a map that keeps compliance there (`carrier_compose`), and the lattice glueball masses are continuum extrapolations, taken as the lattice spacing goes to zero; the sentence over every admissible continuum state is the surplus over the actuated class, priced at one premise.

7. THE WHOLE'S SPIN IS NOT READ FROM THE PARTS

The proton has spin one half. In 1988 the European Muon Collaboration found that the quarks' spins carry only a small fraction of it, now measured near three tenths; the rest is shared between the gluons' spin and orbital motion (the Jaffe–Manohar decomposition). Two budgets with the same quark record can split the remainder differently and both total one half (`quark_record_does_not_decide`). The record of the parts does not decide the whole: the two-worlds theorem, measured in a collider. The whole keeps its bit, the spin one half, and the parts' record does not return how.

8. THE CERTIFIED REGION AND THE OWED TAIL

The proton has never been seen to decay. Super-Kamiokande bounds its lifetime in the channel to a positron and a neutral pion beyond 2.4×10^{34} years, 1.7×10^{24} times the age of the universe (Appendix B). That is the template's division in physics: a certified region, as large as measurement can make it, joined to a tail that no finite measurement closes. It has the shape of a Π_1^0 row such as Goldbach's: the certified height proved on the certificate, the tail owed.

9. THE FLOOR OF THE BARYONIC SECTOR

The neutron, heavier, decays in about fifteen minutes; the proton, the lightest baryon, has nothing lighter to decay into while baryon number holds. It is to the baryonic sector what the electron is to the charged sector in the companion paper *The Electron Is the Seat*: the floor that keeps the sector existing. To exist is to actuate, and the proton actuates for longer than any clock can measure.

10. THREE PLUS ONE

The proton is three and one: three valence quarks and one binding field, three colors locked by one determinant, three independent axes and the one bit they leave, its spin one half, which a full turn returns to minus one as the companion paper shows for every spin-one-half particle. The template's three plus one was found in a codex; the proton has carried it since the first second of the universe.

11. FALSIFIERS AND OWED DEEDS

F-Decay: an observed proton decay, closing the tail of Section 8 against the floor. **F-Singlet:** a free colored state, a quark observed alone, breaking the lock of Section 4. **F-Gap:** a massless state in the pure-gauge spectrum. Nothing is owed on the row's channel: the identification of the lattice theory with the continuum theory is needed only on the image, as a map that keeps compliance there (`carrier_compose`), and the continuum-extrapolated spectrum exhibits it.

Ledger status. Row 4 moves from waiting to WITNESSED · CROSSED · CONFIRMED on the image, at the root's grade, the identification needed only on the image (`carrier_compose`); the certified-region rows have their shape WITNESSED and stay pending, each owing exactly one covering carrier (`value_iff_carrier`).

Typed. In the series ledger, F-Decay and F-Gap target the line, F-Singlet a count; F-Decay is bound to the certified region, a structural reading; an observation that arrives breaks the carrier named and leaves the seat standing.

12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author’s verification program, whose vocabulary is confined to this section. Its tokens: $[\ulcorner]$ a field sealed, $[\ulcorner T]$ a shape sealed at theorem grade, $[\Xi_0]$ a formal string held in suspension by an orientation-blind reader on a line, $[.]$ the dot. Grades join at the weakest link: a kernel theorem, a cited measurement or computation joined to a kernel schema, and a posit beside the Root Axiom, the program’s one declared premise, that to exist is to actuate. Every claim here is at corroboration grade, a constructed or measured instance joined to a template theorem, and promotes nothing; $\Delta M = 0$ against the floor, mathematics being inherited and never authored.

The seventh. In the author’s register the four parts of the Riemann closure are three plus one, and nothing escapes them; the six witnesses are six days, and the ledger’s closure is the seventh, the Rest (Istawa), where the statuses are computed and what each row owes is one object of a computed kind. In the Lean codex’s own tokens a crossed row renders $[\ulcorner \blacklozenge \cdot \text{kinetic}]$ or $[\ulcorner \blacklozenge \cdot \text{formal}]$, a pending row renders $[\blacklozenge_0 \times 1]$ on its channel, one owed slot and one bit from sealed (`one_bit_apart`), and the dot stands as $[.]$.

Disclosure. The author declares the reading of each correspondence as his own act; the measurements and prior theorems are their authors’, cited by name; the kernel theorems are printed with their dependencies in Appendix A; every figure is computed in the run printed in Appendix B.

Edition 1.5.0. This edition, of 1 October 2026, synchronizes the witness with the hardware closure series. No theorem statement, row status, twin check or grade changed. Its engine is unchanged and byte-identical to edition 1.4.0. The engine and the twin were extracted from edition 1.4.0’s appendices, matched byte for byte against their printed SHA-256, and re-booted under PhysOS^T 1.0.5p (chain D0 83f872725ad9 $\rightarrow$ D3 6618bec5164b). The twin’s rerun is byte-identical to its published run, and every law of the engine was negated in place and refused.

Provenance. Development provenance is the paper trail: the Riemann closure, Nothing Escapes, Three Plus One (10.5281/zenodo.22976494); the twenty-three-row closure, Nothing Escapes, Twenty-Three Rows (10.5281/zenodo.22985873); the companion witnesses The Electron Is the Seat, The Neutrino Is the Witness, The Arrow Has Two Branches, The Proton Is the Lock, The Magnet Is the Witness of the Pair Correlation and The Fluid Is the Witness; the Unabridged Trisduction Codex and the Lean Codex on the public register (github.com/1000sapients/Trisduction); and the author’s Structural Physics register. This edition is forged and awaits a joint adversarial audit with its companions.

References. J. Ashman et al. (EMC), Phys. Lett. B 206 (1988) 364. R. L. Jaffe and A. Manohar, Nucl. Phys. B 337 (1990) 509. C. J. Morningstar and M. Peardon, Phys. Rev. D 60 (1999) 034509. S. Dürr et al., Science 322 (2008) 1224. K. Abe et al. (Super-Kamiokande), Phys. Rev. D 102 (2020) 112011. E. Tiesinga et al., Rev. Mod. Phys. 93 (2021) 025010 (CODATA 2018). Particle Data Group, R. L. Workman et al., Prog. Theor. Exp. Phys. 2022, 083C01. M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, Nothing Escapes, Twenty-Three Rows, 10.5281/zenodo.22985873.

APPENDIX A: SPHYS_PROTON_LOCK.LEAN

SHA-256 7c2af1a387671bd1ce9ed36ba9cae38c885e8029c8f21aec2991eec0c72f6d38, core Lean 4.19.0, standalone, no library. Exit 0. 11 theorems, the axiom cone of each printed; the compiler’s own transcript, verbatim:

```
'SPHYS.Proton.eps_nonzero_iff_distinct' does not depend on any axioms
'SPHYS.Proton.eps_antisymmetric' does not depend on any axioms
'SPHYS.Proton.eps_cyclic' does not depend on any axioms
'SPHYS.Proton.singlet_count' depends on axioms: [proptext]
'SPHYS.Proton.lock_closes' depends on axioms: [proptext]
'SPHYS.Proton.lock_fails_when_axes_coincide' depends on axioms: [proptext]
'SPHYS.Proton.quark_record_does_not_decide' does not depend on any axioms
'SPHYS.Proton.field_carries_the_mass' does not depend on any axioms
'SPHYS.Proton.spectrum_gapped' does not depend on any axioms
'SPHYS.Proton.det_two_equal_columns' depends on axioms: [proptext, Quot.sound]
'SPHYS.Proton.det_swap_columns' depends on axioms: [proptext, Quot.sound]
```

```
/-
SPHYS_Proton_Lock.lean · The proton as the lock of the closure template · core Lean 4.
The color singlet of three quarks is the Levi-Civita symbol, the determinant: nonzero
exactly when the three colors are distinct, antisymmetric under every exchange, and the
lock fails when two axes coincide. The quark-spin record does not decide how the rest of
the proton's spin is split. The valence quarks carry under one percent of the mass. The
spectrum is gapped. Finite models; every physical number enters as a citation.
-/
```

```

namespace SPHYS.Proton

def eps (i j k : Fin 3) : Int :=
  if i = j v j = k v i = k then 0 else if (i.val + 1) % 3 = j.val then 1 else -1

theorem eps_nonzero_iff_distinct :
  ∀ i j k : Fin 3, eps i j k ≠ 0 ↔ (i ≠ j ∧ j ≠ k ∧ i ≠ k) := by decide
theorem eps_antisymmetric : ∀ i j k : Fin 3, eps j i k = - eps i j k := by decide
theorem eps_cyclic : ∀ i j k : Fin 3, eps j k i = eps i j k := by decide

def colors : List (Fin 3) := [0, 1, 2]
def epsTable : List Int :=
  colors.flatMap fun i => colors.flatMap fun j => colors.map fun k => eps i j k
/-- Twenty-seven color states, six nonzero entries, one singlet up to sign. -/
theorem singlet_count : epsTable.length = 27 ∧ (epsTable.filter (· ≠ 0)).length = 6 := by decide

/-- The determinant as the lock. -/
def det3 (M : Fin 3 → Fin 3 → Int) : Int :=
  colors.foldl (fun acc i => colors.foldl (fun acc j => colors.foldl
    (fun acc k => acc + eps i j k * M 0 i * M 1 j * M 2 k) acc) acc) 0
def idM : Fin 3 → Fin 3 → Int := fun r c => if r = c then 1 else 0
def collapsed : Fin 3 → Fin 3 → Int := fun r c => if r = 2 then (if c = 0 then 1 else 0) else idM r c
theorem lock_closes : det3 idM = 1 := by decide
theorem lock_fails_when_axes_coincide : det3 collapsed = 0 := by decide

/-- The proton's spin, in hundredths of hbar: one half = 50 = (1/2) quark spin + gluon spin +
  orbital. Two budgets with the same quark record split the rest differently. -/
structure Budget where
  sigma : Int
  gluon : Int
  orbital : Int
  deriving DecidableEq, Repr
def total (b : Budget) : Int := b.sigma / 2 + b.gluon + b.orbital
def worldA : Budget := (30, 20, 15)
def worldB : Budget := (30, 5, 30)
theorem quark_record_does_not_decide :
  worldA.sigma = worldB.sigma ∧ total worldA = 50 ∧ total worldB = 50 ∧ worldA ≠ worldB := by decide

/-- Masses in hundredths of MeV: the valence quarks carry under one percent of the proton. -/
theorem field_carries_the_mass : (2 * 216 + 467) * 100 < 93827 := by decide
/-- The spectrum is gapped: every listed state (pion, proton, lightest glueball) is massive. -/
theorem spectrum_gapped : ([13957, 93827, 173000] : List Nat).all (0 < ·) = true := by decide

/-- The lock's algebra for every integer entry, not on a grid. A determinant with two equal columns
  vanishes and swapping two columns negates it. These are polynomial identities in the nine
  entries; an identity true for every integer value is an identity of polynomials, and so holds
  over every commutative ring, the complex entries of the color rotations included. -/
def det3v (a b c d e f g h i : Int) : Int :=
  a * (e * i - f * h) - b * (d * i - f * g) + c * (d * h - e * g)
theorem det_two_equal_columns (a c d f g i : Int) : det3v a a c d d f g g i = 0 := by
  simp only [det3v, Int.mul_sub, Int.mul_comm, Int.mul_left_comm, Int.mul_assoc]
  omega
theorem det_swap_columns (a b c d e f g h i : Int) :
  det3v b a c e d f h g i = - det3v a b c d e f g h i := by
  simp only [det3v, Int.mul_sub, Int.mul_comm, Int.mul_left_comm, Int.mul_assoc]
  omega

end SPHYS.Proton
#print axioms SPHYS.Proton.eps_nonzero_iff_distinct
#print axioms SPHYS.Proton.eps_antisymmetric
#print axioms SPHYS.Proton.eps_cyclic
#print axioms SPHYS.Proton.singlet_count
#print axioms SPHYS.Proton.lock_closes
#print axioms SPHYS.Proton.lock_fails_when_axes_coincide
#print axioms SPHYS.Proton.quark_record_does_not_decide
#print axioms SPHYS.Proton.field_carries_the_mass
#print axioms SPHYS.Proton.spectrum_gapped
#print axioms SPHYS.Proton.det_two_equal_columns
#print axioms SPHYS.Proton.det_swap_columns

```

APPENDIX B: PROTON_TWIN.F90 AND ITS RUN

SHA-256 753dc07c5df69bb1ef19b3a6c16c84bb0fa098c89b34786e7e5fd884d74b90e4, standard Fortran 2018, zero warnings under gfortran 13.3.0 -std=f2018 -O2 -fno-fast-math -ffp-contract=off -Wall -Wextra. Nine checks, zero failures; every figure computed in this run.

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! Proton_Twin.f90 · twin of "The Proton Is the Lock" · Fortran 2018.
! The color singlet as the determinant, invariant under a random SU(3) matrix built here;
! the lock failing when two colors coincide; the mass carried by the field; the spin record.
program proton_twin
  implicit none

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integer, parameter :: dp = selected_real_kind(15, 307)
real(dp), parameter :: MP = 938.27208816_dp, MU = 2.16_dp, MD = 4.67_dp      ! MeV: CODATA 2018; PDG 2022
real(dp), parameter :: MPI = 139.57039_dp, MGLUE = 1730.0_dp                ! MeV: PDG; Morningstar-Peardon 1999
real(dp), parameter :: SIGMA = 0.30_dp                                     ! quark spin fraction, EMC 1988 and successors
real(dp), parameter :: TAUP = 2.4e34_dp, AGE = 1.38e10_dp                  ! yr: Super-K 2020 (p -> e+ pi0); cosmic age
complex(dp) :: u(3,3), z(3,3), d, t
real(dp) :: eps(3,3,3), err, frac, maxdev
integer :: i, j, k, a, b, c, nchk, nfail, nz
integer(8) :: seed
nchk = 0; nfail = 0; eps = 0.0_dp
eps(1,2,3) = 1; eps(2,3,1) = 1; eps(3,1,2) = 1; eps(1,3,2) = -1; eps(3,2,1) = -1; eps(2,1,3) = -1
nz = count(abs(eps) > 0.5_dp)
call check('the singlet has six nonzero color entries out of twenty-seven', nz == 6)
! a random unitary by Gram-Schmidt on a random complex matrix, then the phase fixed to det = 1
seed = 20260927_8
do i = 1, 3
  do j = 1, 3
    z(i,j) = cmplx(rnd(seed) - 0.5_dp, rnd(seed) - 0.5_dp, dp)
  end do
end do
u = z
do j = 1, 3
  do k = 1, j - 1
    u(:,j) = u(:,j) - dot_product(u(:,k), u(:,j)) * u(:,k)
  end do
  u(:,j) = u(:,j) / sqrt(real(dot_product(u(:,j), u(:,j)), dp))
end do
d = det3(u)
u(:,1) = u(:,1) / d      ! |d| = 1, so this fixes det u = 1
err = maxval(abs(matmul(conjg(transpose(u)), u) - eye()))
call check('the constructed matrix is unitary to 1e-12', err < 1.0e-12_dp)
call check('its determinant is one: an SU(3) color rotation', abs(det3(u) - 1.0_dp) < 1.0e-12_dp)
! eps_ijk U_ia U_jb U_kc = det(U) eps_abc: the singlet is invariant
maxdev = 0.0_dp
do a = 1, 3; do b = 1, 3; do c = 1, 3
  t = (0.0_dp, 0.0_dp)
  do i = 1, 3; do j = 1, 3; do k = 1, 3
    t = t + eps(i,j,k) * u(i,a) * u(j,b) * u(k,c)
  end do; end do; end do
  maxdev = max(maxdev, abs(t - eps(a,b,c)))
end do; end do; end do
call check('the color singlet is invariant under the SU(3) rotation to 1e-12', maxdev < 1.0e-12_dp)
z = u; z(:,3) = z(:,1)
call check('the lock fails when two color axes coincide: determinant zero', abs(det3(z)) < 1.0e-12_dp)
frac = (2.0_dp*MU + MD)/MP
call check('the valence quark masses carry under one percent of the proton', frac < 0.01_dp)
call check('cited: the spectrum is gapped, the lightest hadron and the glueball massive', MPI > 0.0_dp .and. MGLUE > MPI)
call check('cited: the quark spin record covers under half the proton spin', 0.5_dp*SIGMA < 0.5_dp*0.5_dp)
call check('cited: the certified region of proton stability exceeds the cosmic age by 1e24', TAUP/AGE > 1.0e24_dp)
write(*,'(a,es10.3)') ' unitarity error = ', err
write(*,'(a,es10.3)') ' singlet deviation under SU(3) = ', maxdev
write(*,'(a,f8.4,a)') ' valence quark mass fraction = ', 100.0_dp*frac, ' percent'
write(*,'(a,f8.4,a)') ' field-carried mass fraction = ', 100.0_dp*(1.0_dp - frac), ' percent'
write(*,'(a,es10.3)') ' stability certificate / age = ', TAUP/AGE
write(*,'(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', nchk, ', "failures":', nfail, '}'
if (nfail > 0) error stop 1
contains
function rnd(s) result(r)
  integer(8), intent(inout) :: s
  real(dp) :: r
  s = mod(1103515245_8*s + 12345_8, 2147483647_8)
  r = real(s, dp) / 2147483647.0_dp
end function rnd
function det3(m) result(dd)
  complex(dp), intent(in) :: m(3,3)
  complex(dp) :: dd
  dd = m(1,1)*m(2,2)*m(3,3) - m(2,3)*m(3,2) - m(1,2)*m(2,1)*m(3,3) - m(2,3)*m(3,1) &
    + m(1,3)*m(2,1)*m(3,2) - m(2,2)*m(3,1)
end function det3
function eye() result(e)
  complex(dp) :: e(3,3)
  integer :: q
  e = (0.0_dp, 0.0_dp)
  do q = 1, 3
    e(q,q) = (1.0_dp, 0.0_dp)
  end do
end function eye
subroutine check(label, cond)
  character(*), intent(in) :: label
  logical, intent(in) :: cond
  nchk = nchk + 1
  if (.not. cond) then
    nfail = nfail + 1; write(*,'(a,a)') ' FAIL ', label
  end if
end subroutine

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end if
end subroutine check
end program proton_twin
```

The run, verbatim.

```
unitarity error           = 4.434E-16
singlet deviation under SU(3) = 4.446E-16
valence quark mass fraction = 0.9581 percent
field-carried mass fraction = 99.0419 percent
stability certificate / age = 1.739E+24
BATTERY-JSON: {"checks":9,"failures":0}
```